



# Advanced Time Series Analysis and Forecasting

## Chapter 5: Bayesian VAR, factor models and nowcasting



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

## Acronyms used in this material (1/2)

- ▣ ADL-MIDAS: Autoregressive Distributed Lag MIDAS
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BGR: Bańbura–Giannone–Reichlin (large Bayesian VARs, 2010)
- ▣ BIC: Bayesian Information Criterion
- ▣ BVAR: Bayesian Vector AutoRegression
- ▣ CPI: Consumer Price Index
- ▣ CUMFNS: Capacity utilisation, manufacturing (FRED series code)
- ▣ DFM: Dynamic Factor Model
- ▣ DI: Diffusion Index (forecast)
- ▣ DI-AR: Diffusion Index forecast with AutoRegressive lags
- ▣ EM: Expectation–Maximisation (algorithm)
- ▣ EMPL: employment (total private payrolls)
- ▣ ESI: Economic Sentiment Indicator (European Commission)
- ▣ ESS: Effective Sample Size
- ▣ FAVAR: Factor-Augmented Vector AutoRegression
- ▣ FFR: Federal Funds Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ FRED-MD: FRED Monthly Database for macroeconomic research (McCracken–Ng)
- ▣ GDP: Gross Domestic Product
- ▣ GLP: Giannone–Lenza–Primiceri (hierarchical prior selection, 2015)
- ▣ INS: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- ▣ IP: Industrial Production
- ▣ LARGE: the large system of BGR (all series observed over the sample)
- ▣ LLM: Large Language Model
- ▣ MCMC: Markov Chain Monte Carlo

## Acronyms used in this material (2/2)

- MEDIUM: the 20-variable system of BGR
- MIDAS: Mixed DATA Sampling (regression)
- ML: Machine Learning
- MSE: Mean Squared Error
- MSFE: Mean Squared Forecast Error
- NBER: National Bureau of Economic Research
- NIW: Normal-inverse-Wishart (prior)
- NLS: Nonlinear Least Squares
- OLS: Ordinary Least Squares
- PCA: Principal Component Analysis
- RMSE: Root Mean Squared Error
- SE: Standard Error
- SMALL: the 3-variable system of BGR (employment, CPI, federal funds rate)
- TSA: Time Series Analysis
- UNRATE: Unemployment Rate (FRED series code)
- US: United States
- USREC: NBER recession indicator (FRED series code)
- VAR: Vector AutoRegression

## Today's question and route

- ▣ **Question:** how do we forecast and analyse an economy described by a hundred series, some of them published monthly and late, others quarterly?
  - ▶ an unrestricted VAR with 100 variables and 13 lags has more than 130 000 coefficients: the data alone cannot pin them down
- ▣ **Route** of the chapter
  - ▶ the curse of dimensionality; Bayesian inference at the depth we need (conjugacy, Gibbs sampling, MCMC diagnostics)
  - ▶ Minnesota and Normal-inverse-Wishart priors, dummy observations, shrinkage chosen by the marginal likelihood, large BVARs
  - ▶ factor models: principal components, the number of factors, diffusion indexes, FAVAR, dynamic factor models
  - ▶ mixed frequencies and nowcasting: bridge equations, MIDAS, the Kalman filter with a ragged edge, news; Romanian GDP
- ▣ We build on [TSA, Chapter 6](#) (VAR), [TSA, Chapter 10](#) (Kalman filter) and [Chapter 3](#) (structural VAR); Seminar 5 comes before this lecture

## Learning outcomes

- ▣ Derive conjugate posteriors, run a Gibbs sampler and judge its convergence with trace plots, effective sample size and  $\hat{R}$
- ▣ Write the Minnesota prior in natural conjugate form and with dummy observations, and explain each hyperparameter
- ▣ Choose the tightness of a large BVAR by the marginal likelihood or by the in-sample fit rule, and evaluate the forecasts out of sample
- ▣ Estimate approximate factor models by principal components, select the number of factors and build diffusion-index forecasts and a FAVAR
- ▣ Nowcast quarterly GDP from monthly data with bridge equations, MIDAS and a dynamic factor model, and decompose each revision into news

## Reading, data and tools

- Backbone: [Kilian and Lütkepohl \(2017\)](#), Ch. 5 (Bayesian VAR) and Ch. 16 (large VARs, factors); [Karlsson \(2013\)](#); [Stock and Watson \(2016\)](#)
  - ▶ Bayesian background: [Gelman et al. \(2013\)](#); state space: [Durbin and Koopman \(2012\)](#); nowcasting survey: [Bańbura, Giannone, Modugno and Reichlin \(2013\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch\\_05](#)
  - ▶ Minnesota and Normal-inverse-Wishart BVAR, marginal likelihood, PCA with EM, Bai–Ng, FAVAR, Kalman filter and news written in `numpy`
  - ▶ comparison packages: `statsmodels` (DynamicFactorMQ, EM estimation of the nowcasting model) and `bvar` of the Bank of England (GLP hyperparameters)
- Lecture notebook: [open in Google Colab](#)

## Data used in this chapter

| Series                                                                                                             | Source                                                                                                         | Use                  |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------|
| 117 US monthly series in the FRED-MD format (output, labour, housing, orders, money, rates, prices), 1960:1–2026:7 | FRED (St. Louis Fed), one CSV per series; FRED-MD transformation codes <a href="#">McCracken and Ng (2016)</a> | BVAR, factors, FAVAR |
| NBER recession dates; US capacity utilisation, unemployment rate                                                   | FRED (USREC, CUMFNS, UNRATE)                                                                                   | charts, Gibbs        |
| Romania: IP, retail trade, construction, unemployment, six confidence indicators (11 monthly series); euro-area IP | Eurostat (sts_inpr_m, sts_trtu_m, sts_copr_m, une_rt_m, ei_bssi_m_r2)                                          | nowcasting           |
| Romanian real GDP, quarterly, seasonally and calendar adjusted                                                     | Eurostat (namq_10_gdp)                                                                                         | target               |

- All sources are public and need no account or key; FRED-MD codes: 1 level, 2 difference, 4 log, 5 log difference, 6 second log difference
- October 2025: the US household survey and CPI were not collected (federal shutdown); the isolated missing month is interpolated

## Shrinkage was born in Minneapolis



Federal Reserve Bank of Minneapolis, 2010

Photo: Innotata (2010); CC BY-SA 3.0; Wikimedia

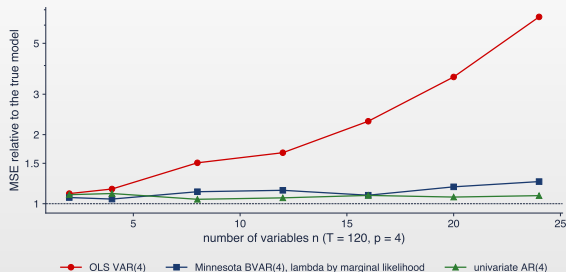
[Commons](#)

- Early 1980s: Litterman forecasts with Bayesian VARs at the Minneapolis Fed [Litterman \(1986\)](#); the “Minnesota prior”
  - ▶ [Doan, Litterman and Sims \(1984\)](#): sum-of-coefficients and conditional projections; [Sims \(1993\)](#), [Sims and Zha \(1998\)](#): dummy initial observations
- 1989–2002: dynamic factor indexes [Stock and Watson \(1989\)](#), generalised dynamic factors [Forni, Hallin, Lippi and Reichlin \(2000\)](#), diffusion indexes [Stock and Watson \(2002a\)](#)
- 2005–2015: FAVAR [Bernanke, Boivin and Elias \(2005\)](#), nowcasting [Giannone, Reichlin and Small \(2008\)](#), large BVARs [Bańbura, Giannone and Reichlin \(2010\)](#), hierarchical priors [Giannone, Lenza and Primiceri \(2015\)](#)
- One idea throughout: when parameters outnumber observations, add information (a prior) or reduce dimension (factors)

## Counting parameters

- ▣ VAR( $p$ ) in  $n$  variables:  $k = np + 1$  regressors per equation,  $nk$  slope coefficients and  $n(n + 1)/2$  covariances
  - ▶  $n = 3, p = 13$ :  $k = 40$ ;  $n = 20$ :  $k = 261$ ;  $n = 102$ :  $k = 1\,327$  per equation
  - ▶ a 10-year window has  $T = 120$  months: with  $k > T$ , OLS does not exist ( $X'X$  is singular)
- ▣ Even with  $k < T$ , the estimation error grows with  $k/T$ : one-step MSE  $\approx \sigma^2(1 + k/T)$  for a correctly specified regression
  - ▶ overfitting: in-sample fit improves, out-of-sample accuracy deteriorates
- ▣ Three remedies: (i) shrinkage (Bayesian priors, ridge, LASSO); (ii) dimension reduction (factors); (iii) variable selection

## Simulation: OLS VAR against a Minnesota BVAR



- True model: stationary VAR(1)  $A = 0.5I + (0.3/n)11'$ ; estimated VAR(4),  $T = 120$ ; one-step MSE of variable 1 relative to the true model; 200 replications

## Interpreting the simulation

- ▣ OLS: the relative MSE climbs to 6.52 at  $n = 24$  ( $k = 97$  regressors for 116 observations)
- ▣ BVAR: 1.25 at  $n = 24$ ; the tightness chosen by the marginal likelihood falls from 0.31 ( $n = 2$ ) to 0.14 ( $n = 24$ )
- ▣ A univariate AR(4) stays near 1.08: the cross-variable information is real but small, and OLS spends it on noise
- ▣ The larger the system, the tighter the prior should be: the theme of the whole chapter [De Mol, Giannone and Reichlin \(2008\)](#)

## Recap: The curse of dimensionality

- ▣ Parameters grow with  $n^2p$ ; observations do not
- ▣ Shrinkage trades a small bias for a large reduction of variance
- ▣ Factors summarise many series by a few common components

## Prior, likelihood, posterior



Pierre-Simon Laplace, who developed inverse probability

Portrait: Paulin Guérin (1838); public domain;  
[Wikimedia Commons](#)

- Bayes:  $p(\theta | y) = p(y | \theta)p(\theta)/p(y)$ , with  $p(y) = \int p(y | \theta)p(\theta) d\theta$ 
  - ▶  $\theta$ : the parameters;  $y$ : the data;  $p(\theta)$ : **prior**;  $p(y | \theta)$ : likelihood;  $p(\theta | y)$ : **posterior**
  - ▶  $p(y)$ : **marginal likelihood**, the evidence for the model and its prior
- Conjugacy: prior and posterior in the same family, so the posterior has a closed form
  - ▶ Normal mean with known variance: Normal; regression coefficients and variance: Normal-inverse-gamma
- Point summaries: posterior mean (quadratic loss), median (absolute loss); credible sets from posterior quantiles

## The Normal update is a precision-weighted average (1/2)

- Data:  $\hat{\beta} \mid \beta \sim N(\beta, s^2)$  with  $s^2 = \sigma^2 / \sum x_t^2$ ; prior:  $\beta \sim N(m_0, v_0)$ 
  - ▶  $\hat{\beta}$ : the OLS estimate of a slope  $\beta$  on regressor  $x_t$ ;  $\sigma^2$ : the error variance, taken as known;  $s^2$ : the sampling variance of  $\hat{\beta}$
  - ▶  $m_0, v_0$ : the prior mean and variance; precision = 1/variance
- Complete the square in  $\beta$ : the posterior is Normal,  $\beta \mid y \sim N(m_1, v_1)$ 
  - ▶ precisions add:  $v_1^{-1} = v_0^{-1} + s^{-2}$
  - ▶ means average:  $m_1 = w m_0 + (1 - w)\hat{\beta}$ , with prior weight  $w = v_0^{-1} / (v_0^{-1} + s^{-2}) \in (0, 1)$
  - ▶  $w$  is large when the prior is tight ( $v_0$  small) or the data are weak ( $s^2$  large)

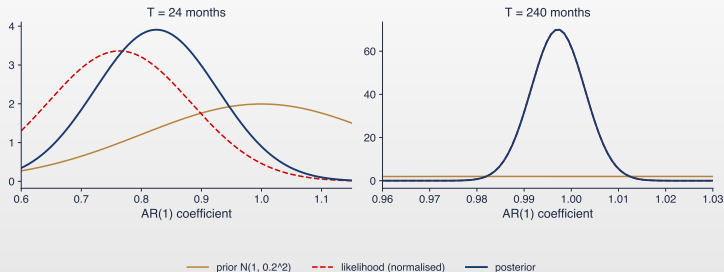
## The Normal update is a precision-weighted average (2/2)

- In regression form:

$$\bar{\beta} = (V_0^{-1} + X'X/\sigma^2)^{-1}(V_0^{-1}m_0 + X'y/\sigma^2)$$

- ▶  $y$  ( $T \times 1$ ),  $X$  ( $T \times k$ ): the data;  $m_0$ ,  $V_0$ : prior mean vector and covariance matrix;  $\bar{\beta}$ : the posterior mean
- ▶ the same weighting: prior precision  $V_0^{-1}$  plus data precision  $X'X/\sigma^2$
- With  $m_0 = 0$  and  $V_0 = \tau^2 I$  this is ridge regression with penalty  $\sigma^2/\tau^2$ 
  - ▶  $\tau^2$ : the prior variance of every coefficient; a small  $\tau^2$  shrinks more towards 0
- As  $T \rightarrow \infty$ ,  $X'X$  grows and the prior weight  $w \rightarrow 0$ : the prior matters in small samples and in large models
- Seminar 5, A1: the update by hand and the empirical-Bayes choice of  $\tau^2$

## Updating an AR(1) coefficient



□ US unemployment rate, monthly, windows ending December 2019; prior  $N(1, 0.2^2)$  (random walk),  $\sigma^2$  at its OLS estimate

## Interpreting the update

- ▣  $T = 24$ : OLS 0.763 (SE 0.119), posterior mean 0.825 (sd 0.102); the prior has weight 26.0%
- ▣  $T = 240$ : OLS 0.997, posterior 0.997; the prior weight falls to 0.1%
- ▣ In a VAR, each equation has hundreds of coefficients but the same  $T$ : the prior keeps the weight it has at  $T = 24$  here

## Marginal likelihood and Bayes factors

- $p(y | M) = \int p(y | \theta, M)p(\theta | M) d\theta$ : the density of the data *before* seeing them
  - ▶  $M$ : a model, i.e. a likelihood together with its prior; the integral averages the likelihood over the prior
  - ▶ prediction-error decomposition:  $\ln p(y) = \sum_t \ln p(y_t | y_{1:t-1})$ , a sum of one-step log scores ([Chapter 1](#));  $y_{1:t-1}$ : the data up to  $t - 1$
  - ▶ it penalises complexity automatically: a diffuse prior spreads probability over data that never occur
- Bayes factor  $B_{12} = p(y | M_1)/p(y | M_2)$ ;  $2 \ln B_{12} > 6$  is “strong” evidence [Kass and Raftery \(1995\)](#)
  - ▶ sensitive to the prior: with an improper prior it is not defined
- Hierarchical use: treat the hyperparameters  $\gamma$  of the prior as parameters and maximise  $p(y | \gamma)$  (empirical Bayes) or put a hyperprior on them [Giannone, Lenza and Primiceri \(2015\)](#)

## Gibbs sampling (1/2)

- When the joint posterior has no closed form but each **full conditional** does: draw  $\theta_1 \mid \theta_2, y$ , then  $\theta_2 \mid \theta_1, y$ , and repeat [Geman and Geman \(1984\)](#), [Gelfand and Smith \(1990\)](#)
  - ▶  $\theta_1, \theta_2$ : two blocks of parameters; full conditional: the distribution of one block given the other and the data
  - ▶ the draws form a Markov chain whose stationary distribution is the posterior
  - ▶ discard a burn-in, then average functions of the draws (ergodic theorem)
- In VARs: stochastic volatility, non-conjugate priors and set-identified SVARs ([Chapter 3](#)) need Gibbs or Metropolis–Hastings steps

## Gibbs sampling (2/2): the linear regression

- Regression  $y = X\beta + e$  with independent priors  $\beta \sim N(b_0, V_0)$ ,  $\sigma^2 \sim IG(a_0, d_0)$ 
  - ▶  $b_0, V_0$ : prior mean and covariance of  $\beta$ ;  $IG(a_0, d_0)$ : inverse gamma with shape  $a_0$  and scale  $d_0$
- Step 1, the coefficients given the variance:  $\beta \mid \sigma^2, y \sim N(\bar{b}, \bar{V})$ 
  - ▶  $\bar{V} = (V_0^{-1} + X'X/\sigma^2)^{-1}$ ,  $\bar{b} = \bar{V}(V_0^{-1}b_0 + X'y/\sigma^2)$ : the Normal update of the previous slides
- Step 2, the variance given the coefficients:  $\sigma^2 \mid \beta, y \sim IG(a_0 + T/2, d_0 + e'e/2)$ 
  - ▶  $e = y - X\beta$ : the residuals at the current draw of  $\beta$ ; the data add  $T/2$  to the shape and half the residual sum of squares to the scale

## MCMC diagnostics

- ▣ **Effective sample size:**  $ESS = M / (1 + 2 \sum_{k \geq 1} \rho_k)$ : the number of independent draws with the same precision
  - ▶  $M$ : the number of draws kept;  $\rho_k$ : the autocorrelation of the draws at lag  $k$
  - ▶ AR(1)-type chain:  $ESS = M(1 - \rho)/(1 + \rho)$ ;  $\rho = 0.9$  keeps about 5% of the draws
- ▣  $\hat{R}$  [Gelman and Rubin \(1992\)](#): several chains from dispersed starts;  $\hat{R} = \sqrt{\hat{V}/W}$ ,  $\hat{V} = \frac{n-1}{n}W + B/n$ 
  - ▶  $n$ : draws per chain;  $W$ : within-chain variance;  $B$ :  $n \times$  the variance of the chain means;  $\hat{V}$ : the pooled variance estimate
  - ▶  $\hat{R} \approx 1$ : the chains agree; current practice: rank-normalised split- $\hat{R} < 1.01$  [Vehtari et al. \(2021\)](#)
- ▣ **Geweke**  $z$ : compares the mean of the first 10% and of the last 50% of a chain, with spectral variances;  $|z| > 2$  signals non-convergence
- ▣ None of them proves convergence; each can reveal a failure

## Gibbs sampling: a bad and a good parametrisation



- Monthly US IP growth on lagged capacity utilisation (mean 79.0%, sd 4.7), 1967–2019,  $T = 635$ ; one-at-a-time Gibbs, four chains, 4 000 draws, burn-in 1 000

## Interpreting the two samplers

- ▣ Raw regressor: the posterior correlation of intercept and slope is  $-0.998$ , so each conditional step moves very little
  - ▶ lag-1 autocorrelation 0.992; ESS = 53 out of 12 000 draws;  $\hat{R} = 1.32$ ; Geweke  $z = -2.27$
- ▣ Centred regressor: correlation  $-0.001$ , ESS = 11 529,  $\hat{R} = 1.00$ , Geweke  $z = 0.01$ 
  - ▶ same model, same posterior of the slope (mean 0.0123); only the sampler changed
- ▣ Lesson: reparametrise or draw correlated blocks jointly; conjugate BVARs avoid the problem by drawing  $B$  in one block

## Recap: Bayesian inference

- Posterior precision = prior precision + data precision; the mean is a weighted average
- The marginal likelihood scores a model with its prior and can choose hyperparameters
- Gibbs sampling needs full conditionals; check ESS,  $\hat{R}$  and Geweke before trusting the draws

## The Minnesota prior

- VAR( $p$ ):  $y_t = c + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$ ;  $(A_l)_{ij}$ : the effect of variable  $j$  at lag  $l$  on variable  $i$ 
  - ▶ prior means [Litterman \(1986\)](#):  $\mathbb{E}[(A_1)_{ii}] = \delta_i$ , all other coefficients 0;  $\delta_i = 1$  (random walk) for persistent series, 0 (white noise) otherwise
  - ▶ prior variances [Bańbura, Giannone and Reichlin \(2010\)](#), eq. (2):  $\text{Var}[(A_l)_{ij}] = \lambda^2/l^2$  if  $j = i$ ,  $\vartheta \lambda^2 \sigma_i^2 / (l^2 \sigma_j^2)$  if  $j \neq i$
- Hyperparameters:  $\lambda$  overall tightness,  $1/l^2$  lag decay,  $\vartheta \in (0, 1]$  cross-variable shrinkage
  - ▶  $\sigma_i^2$ : residual variance of a univariate AR for  $y_i$ , so that  $\sigma_i^2/\sigma_j^2$  fixes units
  - ▶  $\lambda = 0$ : posterior = prior;  $\lambda \rightarrow \infty$ : posterior mean = OLS
- Original version:  $\Sigma$  diagonal and fixed, each equation a separate ridge-type regression with a diffuse constant

## The natural conjugate prior (1/2)

- The VAR as a multivariate regression  $Y = XB + U$ , rows of  $U$  i.i.d.  $N(0, \Sigma)$ 
  - ▶  $Y$  ( $T \times n$ ): the observations;  $X$  ( $T \times k$ ): constant and  $p$  lags;  $B$  ( $k \times n$ ): the coefficients, one column per equation
- Prior [Kadiyala and Karlsson \(1997\)](#):  $\text{vec}(B) \mid \Sigma \sim N(\text{vec}(B_0), \Sigma \otimes \Omega_0)$ ,  $\Sigma \sim IW(\Psi, d)$ 
  - ▶  $\text{vec}$ : stacks the columns;  $\otimes$ : Kronecker product;  $B_0$ : prior mean;  $\Omega_0$  ( $k \times k$ ): prior covariance across regressors
  - ▶  $IW(\Psi, d)$ : inverse Wishart with scale matrix  $\Psi$  and  $d$  degrees of freedom
- Posterior: the same families, with updated parameters
  - ▶  $\bar{\Omega}^{-1} = \Omega_0^{-1} + X'X$ ,  $\bar{B} = \bar{\Omega}(\Omega_0^{-1}B_0 + X'Y)$ : the precision-weighted average again
  - ▶  $\Sigma \mid Y \sim IW(\bar{\Psi}, T + d)$ ,  $\bar{\Psi} = \Psi + \hat{U}'\hat{U} + (\bar{B} - B_0)'\Omega_0^{-1}(\bar{B} - B_0)$ ,  $\hat{U} = Y - X\bar{B}$

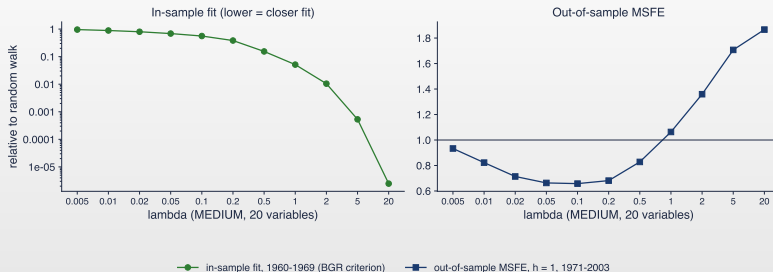
## The natural conjugate prior (2/2)

- ▣ Price of conjugacy: the Kronecker structure gives every equation the same  $\Omega_0$ , so  $\vartheta = 1$ 
  - ▶ Minnesota moments with  $\Omega_0 = \text{diag}(\lambda^2/(l^2\sigma_j^2))$  and  $\mathbb{E}\Sigma = \text{diag}(\sigma_i^2)$  ( $d = n + 2$ ,  $\Psi = \text{diag}(\sigma_i^2)$ )
- ▣ Gain: one  $k \times k$  inversion for all equations; exact draws without MCMC; a closed-form marginal likelihood (▶ [appendix](#))

## Dummy observations

- Theil mixed estimation: add  $T_d$  artificial rows  $(Y_d, X_d)$  and run OLS on  $\begin{pmatrix} Y_d \\ Y \end{pmatrix}$ ,  $\begin{pmatrix} X_d \\ X \end{pmatrix}$ ; each row states a prior belief as if it were data
  - ▶ equivalent to the NIW prior with  $B_0 = (X_d' X_d)^{-1} X_d' Y_d$ ,  $\Omega_0 = (X_d' X_d)^{-1}$  Bańbura, Giannone and Reichlin (2010), eq. (5)
  - ▶ Minnesota block:  $Y_d = \text{diag}(\delta_i \sigma_i) / \lambda$ ,  $X_d = J_p \otimes \text{diag}(\sigma_i) / \lambda$ ,  $J_p = \text{diag}(1, \dots, p)$
- **Sum of coefficients** Doan, Litterman and Sims (1984):  $Y_d = \text{diag}(\bar{y}_i) / \mu$ ,  $X_d = (1_p' \otimes \text{diag}(\bar{y}_i) / \mu, 0)$ 
  - ▶  $\bar{y}_i$ : the mean of the first  $p$  observations of  $y_i$ ;  $1_p$ : a vector of  $p$  ones;  $\mu > 0$ : the tightness of this prior
  - ▶ shrinks  $\Pi = I - \sum_l A_l$  to 0 ("inexact differencing"):  $\mu \rightarrow 0$  gives a VAR in differences without cointegration
- **Dummy initial observation** Sims (1993), Sims and Zha (1998): one row  $y = \bar{y}' / \phi$ ,  $x = (\bar{y}' / \phi, \dots, \bar{y}' / \phi, 1 / \phi)$ 
  - ▶  $\bar{y}$ : the vector of the  $\bar{y}_i$ ;  $\phi > 0$ : its tightness; it pushes towards unit roots or towards a stationary model whose mean is  $\bar{y}$ : allows cointegration (Chapter 4)

## How much shrinkage? In-sample fit against out-of-sample accuracy



- MEDIUM system (20 variables,  $p = 13$ ,  $k = 261$ ), BGR prior, rolling 10-year windows; out-of-sample: one-step MSFE of employment, CPI and the funds rate, 1971–2003, relative to a random walk

## Interpreting the trade-off

- The in-sample fit improves without limit as  $\lambda$  grows: with  $k = 261$  and  $T = 120$ , a loose prior interpolates the data
- Out of sample: U-shape; minimum 0.66 at  $\lambda = 0.10$ ; 0.93 with  $\lambda = 0.005$  (almost the random walk); 1.87 with  $\lambda = 20$
- The shrinkage must be chosen without looking at the evaluation sample: by a fit rule (BGR) or by the marginal likelihood (GLP)

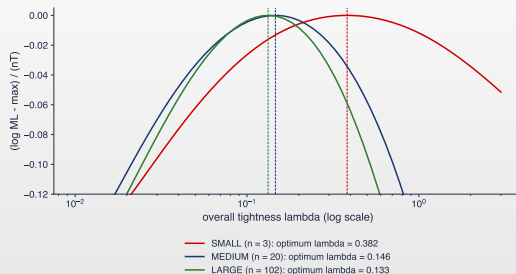
## Recap: Minnesota and NIW priors

- ▣ Minnesota: centre each equation on a random walk or white noise; shrink distant lags and other variables more
- ▣ The natural conjugate form keeps closed forms at the cost of  $\vartheta = 1$
- ▣ Dummy observations implement the prior by OLS and add sum-of-coefficients and initial-observation beliefs

## Giannone, Lenza and Primiceri (2015)

- Treat  $\gamma = (\lambda, \mu, \phi)$  as parameters with hyperpriors:  $p(\gamma | y) \propto p(y | \gamma)p(\gamma)$  Giannone, Lenza and Primiceri (2015)
  - ▶  $p(y | \gamma)$  is known in closed form for the NIW prior with dummies: data plus dummies, minus dummies alone
  - ▶ hyperpriors: Gamma with mode 0.2 and sd 0.4 for  $\lambda$ , mode 1 and sd 1 for  $\mu$  and  $\phi$
- Interpretation: the marginal likelihood is an out-of-sample score of one-step predictive densities (previous section)
  - ▶ so it trades fit against complexity without an evaluation sample
- Use the posterior mode of  $\gamma$  (as here) or integrate  $\gamma$  out by Metropolis–Hastings; GLP find both forecast about as well as factor models

## The marginal likelihood as a function of the tightness



- SMALL ( $n = 3$ ), MEDIUM ( $n = 20$ ), LARGE ( $n = 102$ ),  $p = 13$ , 1960:1–2019:12 ( $T = 707$ );  $\mu = \phi = 1$ ; log ML minus its maximum, divided by  $nT$

## Interpreting the marginal likelihood

- Optimal tightness: 0.382 (SMALL), 0.146 (MEDIUM), 0.133 (LARGE): bigger systems need a tighter prior
- Using the textbook value  $\lambda = 0.2$  costs  $-46.9$  log points for MEDIUM and  $-597.2$  for LARGE;  $\lambda = 1$  costs  $-15571.6$  for LARGE
- Joint mode with the hyperpriors: SMALL  $\mu = 0.20$ ,  $\phi = 0.44$ ; MEDIUM  $\mu = 0.13$ ,  $\phi = 0.44$ : the data ask for tight sum-of-coefficients and initial-observation priors

## Two rules for one hyperparameter

- **BGR fit rule:** choose  $\lambda$  so that the in-sample one-step fit of the key variables equals that of a small OLS VAR Bańbura, Giannone and Reichlin (2010), Section 3
  - ▶  $\text{Fit}(\lambda) = \frac{1}{3} \sum_{i \in I} \text{msfe}_i^{(\lambda)} / \text{msfe}_i^{(0)}$  on 1960–1969; here the target is 0.43
  - ▶  $I$ : the three key variables;  $\text{msfe}_i^{(\lambda)}$ : in-sample one-step mean squared error of variable  $i$  with tightness  $\lambda$ ;  
 $\lambda = 0$ : the prior mean alone
  - ▶ our values: MEDIUM 0.158, LARGE 0.059 (BGR Table 1: 0.108 and 0.035 on their data set)
- **GLP marginal likelihood:** re-estimated on each window; on 10-year windows the median mode is 0.34 (SMALL) and 0.28 (MEDIUM)
  - ▶ 10%–90% range of the MEDIUM mode across windows: [0.23, 0.33]
- The Python package `bvar` of the Bank of England implements the GLP optimisation for the conjugate model; the notebook compares it with our `numpy` code

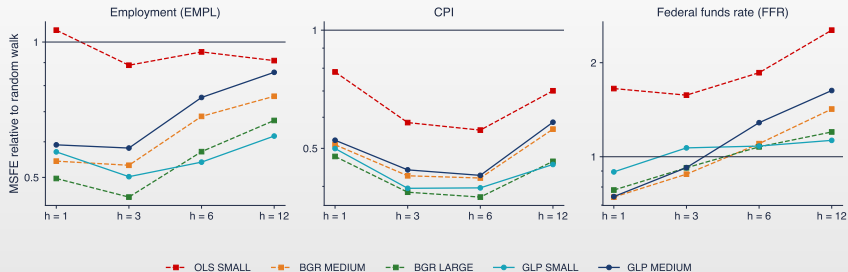
## Recap: Hierarchical priors

- The marginal likelihood chooses the shrinkage; it falls as the system grows
- Sum-of-coefficients and initial-observation priors are chosen in the same way
- A fit rule is a transparent alternative when the evaluation must mimic a published study

## Bańbura, Giannone and Reichlin (2010): the design

- Bańbura, Giannone and Reichlin (2010): can a VAR with 131 variables forecast and identify shocks? Systems SMALL (employment, CPI, funds rate), MEDIUM (20), LARGE (all)
  - ▶  $p = 13$ ; rolling 10-year windows; posterior-mean point forecasts iterated to  $h = 12$ ; benchmark: random walk with drift
  - ▶ Minnesota NIW prior plus sum-of-coefficients with  $\tau = 10\lambda$  (Section 3.3);  $\lambda$  by the fit rule on 1960–1969
- Our data: FRED-MD-format panel; LARGE = the 102 series observed over 1960–2026 (aggregates and components)
  - ▶ MEDIUM: the BGR list, with the S&P 500, the effective exchange rate and nonborrowed reserves replaced by the Baa yield, the 3-month bill rate and building permits (not available for 1960–2026 or negative after 2008)
- Evaluation: targets 1971–2003 (the paper), 2004–2019 and 2004–2026 (extensions); 678 forecast origins; GLP SMALL and MEDIUM added

## Forecast accuracy, 1971–2003



- MSFE relative to the random walk with drift (log scale; below 1 = better), employment and CPI in  $100 \times \log$  levels, funds rate in percent

ATS\_ch5\_large\_bvar

## Interpreting the forecast comparison

- ▣  $h = 1$ , employment: OLS SMALL 1.06, BGR MEDIUM 0.54, LARGE 0.50 (BGR Table 1: 1.14, 0.54, 0.46)
  - ▶ CPI: 0.78, 0.51, 0.48; funds rate: 1.65, 0.74, 0.78
- ▣ Adding variables with more shrinkage helps: LARGE beats SMALL for every variable at  $h \leq 6$
- ▣ GLP SMALL is nearly as good as the large systems for employment and CPI (0.57, 0.50): most of the gain comes from the prior, not only from the variables
- ▣ The funds rate beyond 3 months is hard for all models (LARGE 1.20 at  $h = 12$ )

## After 2003: the extension and March 2020

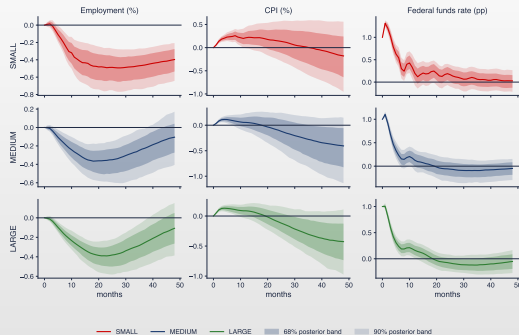
| Model      | EMPL, 1 | EMPL, 12 | CPI, 1 | CPI, 12 | FFR, 1 | FFR, 12 |
|------------|---------|----------|--------|---------|--------|---------|
| OLS SMALL  | 0.38    | 1.08     | 1.26   | 2.43    | 0.98   | 3.51    |
| BGR MEDIUM | 0.20    | 0.66     | 0.97   | 6.49    | 0.55   | 1.91    |
| BGR LARGE  | 0.19    | 0.67     | 0.92   | 3.73    | 0.45   | 1.79    |
| GLP SMALL  | 0.24    | 0.33     | 0.93   | 1.07    | 0.61   | 0.67    |
| GLP MEDIUM | 0.20    | 0.28     | 0.95   | 2.55    | 0.57   | 0.91    |

- Targets 2004:1–2019:12, relative MSFE (random walk with drift = 1)
- Including 2020–2026, the 12-month employment MSFE explodes: BGR MEDIUM 500.06, GLP SMALL 8.64: windows containing April 2020 fit the collapse as dynamics
- Remedy: scale the residual variance of the pandemic months by estimated factors [Lenza and Primiceri \(2022\)](#), or stochastic volatility (next section)

## A monetary policy shock in a large VAR

- Recursive scheme of Bańbura, Giannone and Reichlin (2010), Section 4:  $y_t = (X_t, r_t, Z_t)$ , slow variables  $X_t$  (real activity, prices), the funds rate  $r_t$ , fast variables  $Z_t$  (money, rates, exchange rates)
  - ▶ only the position of  $r_t$  matters (Chapter 3): no ordering inside the blocks is needed
- Bayesian bands: for each draw  $(B, \Sigma)$  from the NIW posterior, compute the Cholesky impact column and the responses; report posterior quantiles
  - ▶ 500 draws, 100 bp shock, sample 1961–2002,  $p = 13$ ; SMALL with a nearly flat prior
- These are the posterior bands Chapter 3 announced: pointwise credible intervals, conditional on the identifying assumption

## Responses to a 100 bp funds-rate shock



- Posterior medians with 68% and 90% bands; rows: SMALL, MEDIUM, LARGE; employment and CPI in percent

Q.ATS\_ch5\_large\_bvar

## Interpreting the responses

- Employment: trough  $-0.49\%$  (SMALL, month 28),  $-0.39\%$  (LARGE, month 23); after four years LARGE is back to  $-0.11\%$  (band  $[-0.39, 0.15]$ )
- CPI: the price puzzle shrinks from  $0.26\%$  (SMALL) to  $0.13\%$  (LARGE); after four years LARGE gives  $-0.43\%$  (band  $[-0.97, 0.11]$ )
- As in BGR: more information makes the employment response less persistent and the price response more plausible
- The bands do not widen with 102 variables: the prior pays for the extra coefficients

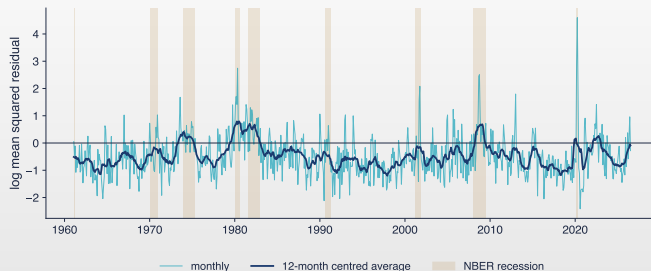
## Recap: Large BVARs

- ▣ With shrinkage that grows with the system, a 102-variable VAR forecasts better than a 3-variable OLS VAR
- ▣ The 1971–2003 results of BGR replicate on today's data; after 2020 the homoskedastic BVAR fails
- ▣ Large information sets reduce the price puzzle of small monetary VARs

## Large BVARs with stochastic volatility

- **Common volatility** [Carriero, Clark and Marcellino \(2016\)](#):  $u_t = \sqrt{f_t} \varepsilon_t$ ,  $\varepsilon_t \sim N(0, \Sigma)$ ,  $\ln f_t = \phi \ln f_{t-1} + \nu_t$ 
  - ▶  $u_t$ : the VAR residual;  $f_t > 0$ : a volatility factor common to all equations;  $\phi$ : its persistence;  $\nu_t$ : its shock
  - ▶ given  $f_{1:T}$ , the model is a weighted conjugate BVAR (rows divided by  $\sqrt{f_t}$ );  $f_t$  is drawn by a Kim–Shephard–Chib step
- **Variable-specific volatilities** [Carriero, Clark and Marcellino \(2019\)](#), [Clark \(2011\)](#):  $\Sigma_t = A^{-1} H_t A^{-1'}$ ,  $H_t$  diagonal with log-AR volatilities [Primiceri \(2005\)](#)
  - ▶  $A$ : lower triangular with unit diagonal (the contemporaneous relations);  $H_t$ : the time-varying variances of the orthogonalised shocks
  - ▶ conjugacy is lost; the triangular algorithm draws the equations one at a time and makes  $n = 100$  feasible
- **Gains**: mainly in density forecasts (calibrated intervals) and in robustness to outliers such as 2020

## A common volatility factor in the BVAR residuals



□ MEDIUM BVAR (GLP hyperparameters), 1960–2026: log of the cross-sectional mean of the squared standardised residuals

ATS\_ch5\_large\_bvar

## Interpreting the volatility proxy

- Common movements: the residual variance rises in every recession and falls in the Great Moderation (1960–1984 is 1.28 times 1985–2007)
- April 2020 is 176 times the median month: a homoskedastic Gaussian BVAR treats it as an ordinary observation
- This is the case for the common-volatility model (one extra state) before the full variable-specific model; [Chapter 6](#) gives the filtering tools

## The approximate factor model

- $x_{it} = \lambda_i' F_t + e_{it}$ ,  $i = 1, \dots, N$ ,  $t = 1, \dots, T$ ;  $F_t$ :  $r$  common factors,  $\lambda_i$ : loadings,  $e_{it}$ : idiosyncratic
  - ▶ *approximate* Chamberlain and Rothschild (1983): the  $e_{it}$  may be weakly correlated across  $i$  and over  $t$ , provided the correlation is bounded as  $N \rightarrow \infty$
  - ▶  $\Lambda$  ( $N \times r$ ): the stacked loadings;  $X$  ( $T \times N$ ): the panel;  $\Sigma_\Lambda$ : a positive definite limit
  - ▶ pervasiveness:  $\Lambda' \Lambda / N \rightarrow \Sigma_\Lambda > 0$ , every factor affects many series; the  $r$  largest eigenvalues of  $XX'$  grow with  $N$ , the others do not
- Identification:  $\Lambda F_t = (\Lambda H)(H^{-1} F_t)$  for any invertible  $H$ : only the factor space is identified
  - ▶ normalisation for PCA:  $F' F / T = I_r$  and  $\Lambda' \Lambda$  diagonal
- Static form of a dynamic model: lags of the dynamic factors are stacked in  $F_t$  Stock and Watson (2016); the generalised dynamic model works in the frequency domain Forni, Hallin, Lippi and Reichlin (2000)

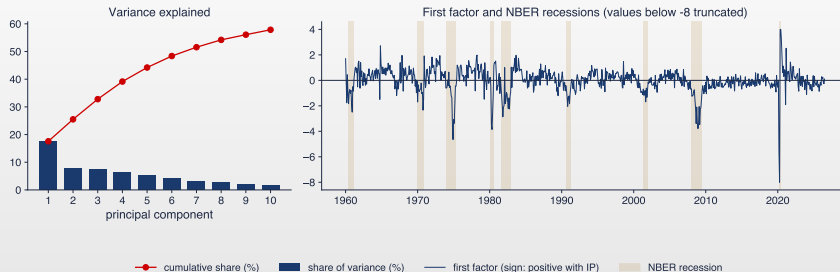
## Principal components

- Least squares:  $\min_{F, \Lambda} \sum_{i,t} (x_{it} - \lambda'_i F_t)^2$  with  $F'F/T = I$ :  $\hat{F} = \sqrt{T} \times$  the first  $r$  eigenvectors of  $XX'$ ,  $\hat{\Lambda} = X'\hat{F}/T$ 
  - ▶ standardise each series first: PCA is not scale invariant
- Consistency [Stock and Watson \(2002a\)](#), [Bai \(2003\)](#):  $\hat{F}_t \rightarrow HF_t$  as  $N, T \rightarrow \infty$  at rate  $\min(\sqrt{N}, \sqrt{T})$ 
  - ▶  $H$ : an invertible  $r \times r$  rotation; PCA recovers the factors up to this rotation
  - ▶ if  $\sqrt{T}/N \rightarrow 0$ , the estimated factors can be used as regressors as if observed (no generated-regressor correction)
- Missing values and mixed start dates: EM [Stock and Watson \(2002b\)](#), [McCracken and Ng \(2016\)](#): fill with 0, run PCA, refill the missing cells with  $\hat{\lambda}'_i \hat{F}_t$ , iterate

## The FRED-MD database

- ▣ [McCracken and Ng \(2016\)](#): about 130 US monthly series since 1959, updated monthly, with a transformation code per series and vintages for real-time work
  - ▶ eight groups: output and income, labour market, housing, consumption and orders, money and credit, interest and exchange rates, prices, stock market
  - ▶ outliers: an observation more than 10 interquartile ranges from the median is set to missing (3 cells here)
- ▣ Our panel: 117 FRED-MD series that FRED distributes (stock-market series are not), 1960:1–2026:7, 4.5% missing cells
  - ▶ the official file (`current.csv` of the St. Louis Fed) can replace it in the notebook with one line
- ▣ Current vintage only: the results are pseudo out of sample, not real time [Croushore and Stark \(2001\)](#)

## Factors of the FRED-MD panel



- Left: share of the variance of the standardised panel explained by each principal component; right: the first factor (EM, 8 factors), NBER recessions shaded

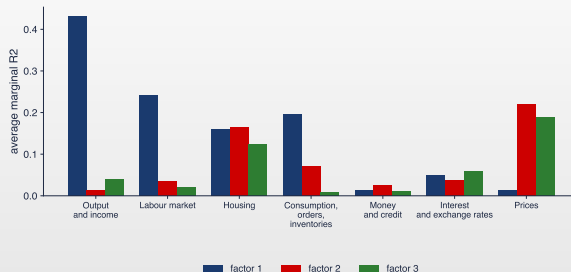
## Interpreting the factors

- The first component explains 17.6%, the second 7.9%; five explain 44.2%, eight 54.2%
- The first factor is a real activity index: correlation  $-0.58$  with the NBER recession indicator; its minimum is April 2020
- No component dominates: macro panels have a few strong factors and a long tail of weak ones, which makes the number of factors a real question

## How many factors? Bai and Ng (2002)

- $V(k) = (NT)^{-1} \sum_{i,t} (x_{it} - \hat{\lambda}_i' \hat{F}_t)^2$  after  $k$  components; it falls with  $k$ , so it needs a penalty Bai and Ng (2002)
  - ▶  $IC_{p1}(k) = \ln V(k) + k \frac{N+T}{NT} \ln \frac{NT}{N+T}$ ,  $IC_{p2}(k) = \ln V(k) + k \frac{N+T}{NT} \ln C_{NT}^2$ ,  $C_{NT}^2 = \min(N, T)$
  - ▶  $IC_{p3}(k) = \ln V(k) + k \ln C_{NT}^2 / C_{NT}^2$ ; choose the  $k$  that minimises the criterion; the penalty grows with  $k$  and vanishes as  $N, T \rightarrow \infty$
  - ▶  $PC_p$  versions:  $V(k) + k \hat{\sigma}^2 g(N, T)$  with  $\hat{\sigma}^2 = V(k_{\max})$ ,  $g(N, T)$  the same penalties: depend on  $k_{\max}$ , the largest  $k$  tried
- Our panel ( $k_{\max} = 15$ ):  $IC_{p1} = 8$ ,  $IC_{p2} = 8$ ,  $IC_{p3} = 10$ ,  $PC_{p1} = 13$ ,  $PC_{p2} = 11$ 
  - ▶ the criteria disagree when the eigenvalues decline slowly; report the choice and its sensitivity
- For forecasting, the number of factors in the regression is chosen separately (by BIC), as in diffusion-index forecasts

## The economic content of the factors



□ Average marginal  $R^2$  of the regression of each standardised series on one factor, by group (as in McCracken and Ng 2016)

Q.ATS\_ch5\_factors

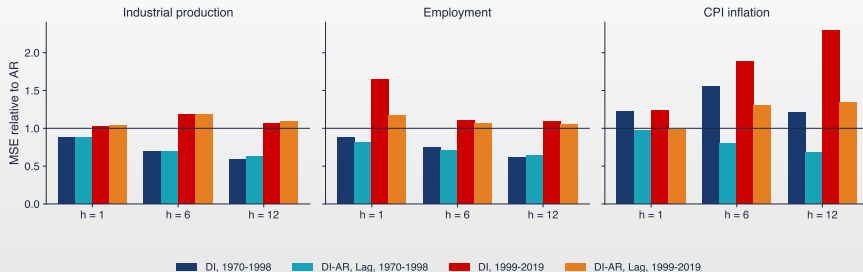
## Interpreting the marginal $R^2$

- Factor 1 is real activity:  $R^2$  0.43 for output, 0.24 for labour; the best single series has 0.78
- Factors 2 and 3 load on prices (0.22, 0.19) and housing (0.16, 0.12)
- Interest rates load weakly on the first three factors: monetary information sits in later factors or in the rates themselves, the motivation for the FAVAR
- Labels are a rotation choice: the factor space, not each factor, is identified

## Diffusion-index forecasts: Stock and Watson (2002)

- **Stock and Watson (2002b):**  $y_{t+h}^h = \alpha_h + \beta_h' \hat{F}_t + \sum_{j=0}^p \gamma_{hj} z_{t-j} + \varepsilon_{t+h}$  (direct forecast, one equation per  $h$ )
  - ▶  $y_{t+h}^h$ : the target over the next  $h$  months, annualised;  $\hat{F}_t$ : the estimated factors;  $z_t$ : the target's own current growth;  $\alpha_h, \beta_h, \gamma_{hj}$ : coefficients estimated by OLS for each  $h$
  - ▶ real series:  $y_{t+h}^h = \frac{1200}{h} \ln(Y_{t+h}/Y_t)$ ; prices:  $\frac{1200}{h} \ln(P_{t+h}/P_t) - 1200 \ln(P_t/P_{t-1})$  ( $Y_t, P_t$ : levels)
  - ▶ DI: factors only,  $k \in \{1, \dots, 12\}$  by BIC; DI-AR, Lag: factors and  $p \in \{0, \dots, 5\}$  lags by BIC; benchmark: AR with BIC lags
- Recursive: factors re-estimated at each origin by EM on the data available then; origins from 1970:1
  - ▶ evaluation 1970–1998 (the paper) and 1999–2019 (extension), by forecast origin
- Targets: industrial production, payroll employment, CPI inflation;  $h = 1, 6, 12$  months

## Diffusion-index forecasts against an AR



□ MSE relative to the AR forecast with BIC lags (below 1 = better)

## Interpreting the diffusion indexes

- ▣ 1970–1998: large gains for real activity at  $h = 12$ : IP 0.58, employment 0.62; for inflation only with own lags (DI-AR, Lag 0.68)
- ▣ 1999–2019: the gains vanish (IP 1.07, employment 1.09, inflation 1.35): the Great Moderation made activity less predictable from the panel
- ▣ BIC selects a median of 7 factors (10%–90%: 5–12) for DI-AR, Lag at  $h = 12$
- ▣ Bayesian shrinkage and principal components give highly correlated forecasts in large panels [De Mol, Giannone and Reichlin \(2008\)](#): two ways to use the same information

## Recap: Factor models

- A few factors summarise a hundred series; PCA estimates the factor space consistently as  $N, T \rightarrow \infty$
- Bai–Ng criteria choose the number of factors; check their sensitivity
- Diffusion indexes helped strongly before 1999 and much less after: evaluate on more than one period

## FAVAR: Bernanke, Boivin and Elias (2005)

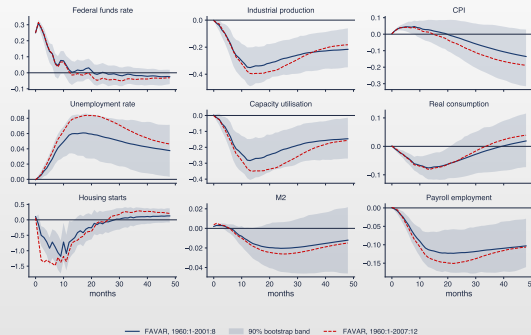


Ben Bernanke, 2008

Photo: Federal Reserve (2008); public  
domain; Wikimedia Commons

- Bernanke, Boivin and Elias (2005):  $\begin{pmatrix} F_t \\ Y_t \end{pmatrix} = \Phi(L) \begin{pmatrix} F_{t-1} \\ Y_{t-1} \end{pmatrix} + v_t$ ,  
 $X_t = \Lambda^f F_t + \Lambda^y Y_t + e_t$ ,  $Y_t$  = funds rate
  - ▶  $F_t$ :  $K$  unobserved factors;  $X_t$ : the large panel;  $\Lambda^f$ ,  $\Lambda^y$ : loadings;  $\Phi(L)$ : a lag polynomial;  $v_t$ ,  $e_t$ : errors
  - ▶ the VAR sees the information of 100 series through  $K$  factors; responses are available for every series in  $X_t$
- Two-step estimation (Section III): principal components  $\hat{C}_t$  of  $X_t$ ; slow factors from slow-moving series; remove  $Y_t$ :  $\hat{F}_t = \hat{C}_t - \hat{b}_Y Y_t$ 
  - ▶  $K = 3$ , VAR(13), 1960:1–2001:8 (ours:  $T = 500$ ,  $N = 102$  balanced series, 76 slow); Cholesky with  $Y_t$  last; 25 bp shock
- Prices and money enter as monthly growth rates; responses are cumulated to levels

## FAVAR responses to a 25 bp monetary policy shock



- Levels in percent (rates and unemployment in pp); 90% residual-bootstrap bands, factors treated as data; dashed: sample to 2007:12

## Interpreting the FAVAR

- ▣ Real activity falls with a lag: IP trough  $-0.35\%$  after 13 months, capacity utilisation  $-0.28$  pp, payrolls  $-0.12\%$ , unemployment up by  $0.061$  pp
- ▣ Housing starts react first and most ( $-1.18\%$  after 9 months): the interest-sensitive sector
- ▣ The price puzzle is small (at most  $0.046\%$ , month 8); CPI ends at  $-0.14\%$  after 48 months (band  $[-0.32, 0.03]$ ); to 2007:  $-0.19\%$
- ▣ Seminar 5, B4: with  $K = 1$  or  $K = 5$  the puzzle returns; the choice of  $K$  is an identifying assumption, not a detail

## The dynamic factor model

- Measurement:  $x_t = \Lambda f_t + e_t$ ,  $e_t \sim N(0, \Psi)$ ,  $\Psi$  diagonal; transition:  $f_t = A_1 f_{t-1} + \dots + A_q f_{t-q} + u_t$ ,  $u_t \sim N(0, Q)$ 
  - ▶  $x_t$ : the  $N$  observed series;  $f_t$ : the  $r$  factors;  $\Lambda$ : loadings;  $e_t$ : idiosyncratic errors with diagonal covariance  $\Psi$ ;  $A_j$ : factor VAR( $q$ ) matrices;  $Q$ : covariance of the factor shocks  $u_t$
  - ▶ a state-space model (TSA, Chapter 10; Chapter 6): state  $s_t = (f_t', \dots, f_{t-m}')'$ ,  $m \geq q - 1$  lags; Kalman filter and smoother
- **Two-step** Doz, Giannone and Reichlin (2011): PCA for  $\hat{f}_t$ , OLS for  $\Lambda$  and  $\Psi$ , a VAR on  $\hat{f}_t$  for  $A$  and  $Q$ , then the Kalman smoother re-estimates  $f_t$ 
  - ▶ consistent for large  $N$  and  $T$  even though  $\Psi$  diagonal is misspecified
- **Quasi-maximum likelihood by EM** Doz, Giannone and Reichlin (2012), Bańbura and Modugno (2014): E-step = Kalman smoother, M-step = regressions on smoothed moments; handles any pattern of missing data and idiosyncratic AR(1) terms

## Missing data are not a problem for the Kalman filter

- At time  $t$ , keep only the observed rows:  $x_t^o = W_t x_t$ ,  $\Lambda_t = W_t \Lambda$ ,  $\Psi_t = W_t \Psi W_t'$  with  $W_t$  a selection matrix
  - ▶ if nothing is observed, the update step is skipped and the prediction is carried forward
- **Ragged edge**: at the end of the sample each series stops at a different month (publication lags); the filter uses whatever is there
  - ▶ the same mechanism handles series that start late (services confidence from 2002) and quarterly series observed every third month
- This is why nowcasting is a filtering problem, and why [Chapter 6](#) develops the filter in full

## The nowcasting problem



Lucrezia Reichlin, 2013

Photo: Niccolò Caranti (2013); CC BY-SA 3.0;

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- **Nowcasting**: estimating the present and the recent past of a low-frequency variable (GDP) from timelier high-frequency data [Giannone, Reichlin and Small \(2008\)](#), [Bańbura, Giannone, Modugno and Reichlin \(2013\)](#)
  - ▶ backcast (last quarter, not yet published), nowcast (this quarter), forecast (next quarter)
- Three difficulties: mixed frequencies, many indicators, asynchronous releases (the ragged edge)
  - ▶ surveys are out at the end of the month; industrial production more than a month later; GDP weeks after the quarter
- Tools: bridge equations [Baffigi, Golinelli and Parigi \(2004\)](#), MIDAS [Ghysels, Santa-Clara and Valkanov \(2006\)](#), [Ghysels, Sinko and Valkanov \(2007\)](#), mixed-frequency DFM [Mariano and Murasawa \(2003\)](#), [Bańbura and Modugno \(2014\)](#) and mixed-frequency VARs [Schorfheide and Song \(2015\)](#)

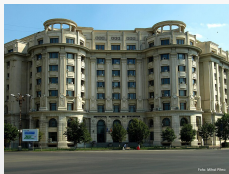
## Bridge equations and MIDAS

- **Bridge**: forecast the missing months of each indicator (AR), aggregate to the quarter, regress GDP on the quarterly aggregates [Baffigi, Golinelli and Parigi \(2004\)](#)
  - ▶ simple and transparent; the error of the monthly AR forecasts passes into the nowcast
- **MIDAS** [Ghysels, Santa-Clara and Valkanov \(2006\)](#):  $y_\tau = \beta_0 + \beta_1 \sum_{j=0}^{K-1} w_j(\theta) x_{m_\tau-j} + \rho y_{\tau-s} + \varepsilon_\tau$ , with the latest monthly value  $x_{m_\tau}$ 
  - ▶  $\tau$ : the quarter;  $m_\tau$ : its latest available month;  $x$ : a monthly indicator with  $K$  lags;  $w_j(\theta)$ : weights summing to 1;  $y_{\tau-s}$ : lagged GDP growth
  - ▶ exponential Almon:  $w_j(\theta) = e^{\theta_1 j + \theta_2 j^2} / \sum_k e^{\theta_1 k + \theta_2 k^2}$ ; two parameters for  $K$  lags, estimated by NLS
  - ▶ MIDAS with leads [Andreou, Ghysels and Kourtellis \(2013\)](#), [Clements and Galvão \(2008\)](#): one regression per information set; U-MIDAS leaves  $w_j$  unrestricted when  $K$  is small [Forni, Marcellino and Schumacher \(2015\)](#)
- Neither uses the joint dynamics of the indicators; the DFM does

## Linking months and quarters: Mariano and Murasawa (2003)

- Let  $Y_t^M$  be a latent monthly level and the quarterly GDP the 3-month average, observed in the third month
  - ▶ geometric-mean approximation:  $\ln Y_t^Q \approx \frac{1}{3}(\ln Y_t^M + \ln Y_{t-1}^M + \ln Y_{t-2}^M)$
  - ▶ quarterly growth  $y_t^Q = \ln Y_t^Q - \ln Y_{t-3}^Q = \frac{1}{3}(y_t + 2y_{t-1} + 3y_{t-2} + 2y_{t-3} + y_{t-4})$ ,  $y_t$  = monthly growth  
Mariano and Murasawa (2003)
- In the DFM:  $y_t^Q = \lambda_y' \sum_{j=0}^4 w_j f_{t-j} + e_t^Q$ ,  $w = (1, 2, 3, 2, 1)/3$ : the state must hold five lags of  $f_t$ 
  - ▶  $\lambda_y$ : the loadings of GDP growth;  $w_j$ : the Mariano–Murasawa weights;  $e_t^Q$ : the idiosyncratic error of GDP
  - ▶ GDP is a monthly variable observed every third month: missing in months 1 and 2 of each quarter
- Seminar 5, A7: the weights by hand

## Case study: nowcasting Romanian GDP



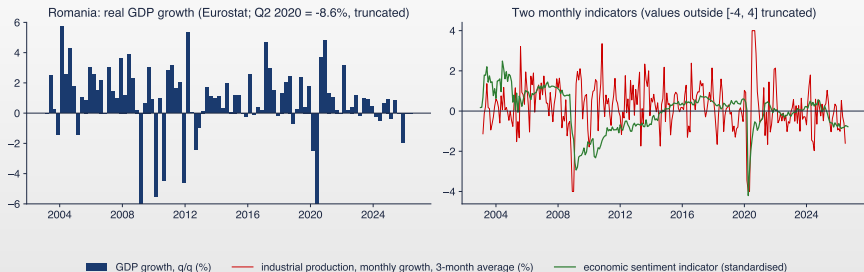
Institutul Național de Statistică, Bucharest

Photo: Dan Mihai Pitea (2009); CC BY-SA 3.0;

[Wikimedia Commons](#)

- Target: q/q growth of real GDP (Eurostat), 94 quarters to 2026Q2; monthly panel: 11 series from 2003 (Eurostat)
  - ▶ hard data in monthly growth: IP, retail, construction, euro-area IP; change of unemployment; levels of six confidence balances
- Stylised release lags at the end of each month: surveys 0, unemployment 1, hard data 2 months; GDP 2 months after the quarter
  - ▶ current-vintage data: pseudo real time (no revisions)
- Models: AR(1), bridge (IP, retail, ESI), MIDAS (IP, ESI;  $K = 6$ ), DFM with two factors (two-step and EM)

## Romanian GDP and two monthly indicators

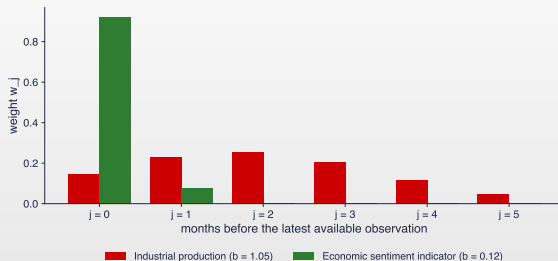


□ Quarterly GDP growth (2020Q2: -8.6%, truncated); IP growth (3-month average) and the standardised ESI

## Interpreting the Romanian data

- ▣ Quarterly growth is volatile: 2009–2012 swings of  $\pm 5$  pp and the 2020 collapse dominate the variance
- ▣ The ESI (3-month average) has correlation 0.44 with GDP growth before 2020: informative, far from perfect
- ▣ Latest published quarter: 2026Q2, growth 0.00%; the nowcast target is 2026Q3

## MIDAS weights for Romania



□ ADL-MIDAS with exponential Almon weights at the end of the third month of the quarter, 2003–2026

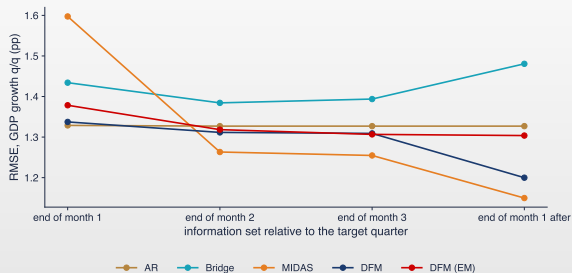
## Interpreting the MIDAS weights

- ▣ IP: hump-shaped weights peaking at lag 2 (0.25): the months of the quarter itself matter, as the Mariano–Murasawa weights predict
- ▣ ESI: weight 0.92 on the latest month: a level indicator already summarises the past
- ▣ Lagged GDP growth has coefficient  $-0.28$ : the noisy q/q series mean-reverts

## Pseudo-real-time evaluation

- For each quarter 2013Q1–2026Q2, nowcasts at the ends of months 1, 2, 3 of the quarter and of month 1 after it
  - ▶ all parameters re-estimated on the data available at that date; 2020Q2–Q3 excluded from the RMSE (also reported with them)
- The evaluation sample starts in 2013 because the 2009–2012 swings are larger than any model error; from 2010 the RMSEs rise to about 2 pp for every model
  - ▶ standard deviation of the target 2013–2026 (without 2020Q2–Q3): 1.33 pp
- Question: does the RMSE fall as the quarter unfolds, and by how much relative to an AR?

## Nowcast accuracy by information set



RMSE of the nowcasts of  $q/q$  GDP growth, 2013Q1–2026Q2 without 2020Q2–Q3

ATS\_ch5\_nowcast

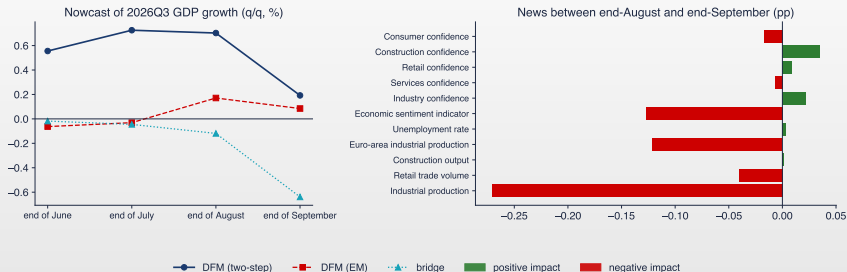
## Interpreting the nowcast accuracy

- ▣ AR: 1.33 pp throughout; DFM: 1.34 (month 1) to 1.20 (month 1 after); MIDAS: 1.60 to 1.15
- ▣ Bridge does not beat the AR (1.39); the EM estimate of the DFM is close to the AR (1.31 at month 3)
- ▣ Diebold–Mariano DFM against AR at month 3:  $t = -0.30$ ,  $p = 0.76$ : the gain is not significant on 54 quarters
- ▣ With 2020: AR 1.90, DFM 1.30 at month 1 after: the monthly data catch the collapse, the AR cannot

## News and revisions: Bańbura and Modugno (2014)

- Information sets  $\Omega_v \subset \Omega_{v+1}$ ; the new releases  $x_j$ ,  $j \in J_{v+1}$ ; the **news** is  $l_j = x_j - \mathbb{E}[x_j \mid \Omega_v]$ , not  $x_j$ 
  - ▶ with fixed parameters the revision of the nowcast of  $y$  is a weighted sum of the news:  
 $\mathbb{E}[y \mid \Omega_{v+1}] - \mathbb{E}[y \mid \Omega_v] = \mathbb{E}[yI']\mathbb{E}[II']^{-1}I = \sum_j w_j l_j$  Bańbura and Modugno (2014)
  - ▶  $I$ : the vector of news  $l_j$ ;  $w_j$ : the weight of release  $j$
- In the DFM:  $l_j = \lambda_j'(f_{t_j} - \hat{f}_{t_j|v}) + e_j$ , so  $\mathbb{E}[II'] = HP_{|v}H' + \Psi_J$  and  $\mathbb{E}[yI'] = z_y'P_{|v}H'$ 
  - ▶  $\lambda_j$ : the loadings of release  $j$ ;  $t_j$ : its reference month;  $\hat{f}_{t_j|v}$ : the factor estimate given  $\Omega_v$ ;  $H$ : the stacked  $\lambda_j'$ ;  $\Psi_J$ : their idiosyncratic variances;  $z_y$ : the loading of the target
  - ▶  $P_{|v}$ : the covariance of the stacked state given  $\Omega_v$ , from the Kalman filter
- A large release that was expected moves nothing; a small surprise in a heavily weighted series moves a lot; correlated releases share their weight

## Nowcasting 2026Q3, and where the revision came from



- Left: nowcasts at the ends of June–September 2026; right: news decomposition of the two-step DFM nowcast between the August and September information sets

## Interpreting the news

- DFM nowcast of 2026Q3: 0.70% at the end of August, 0.19% at the end of September; the 11 new releases add up to  $-0.51$  pp exactly
- The July IP release was  $-3.7$  pp below the model's expectation; with weight 0.073 it cut the nowcast by  $-0.27$  pp; ESI  $-0.13$ , euro-area IP  $-0.12$
- Other models at the end of September: EM DFM 0.08%, bridge  $-0.64\%$ , AR 0.76%: the first official estimate for 2026Q3 will judge them
- The news decomposition turns a number into a story that a central bank can check release by release

## Recap: Nowcasting

- ▣ Nowcasting = filtering a mixed-frequency system with a ragged edge
- ▣ Bridge, MIDAS and DFM share the Mariano–Murasawa link between months and quarters
- ▣ For Romania, monthly data improve the nowcast late in the quarter; the gain is modest and not significant
- ▣ Every revision decomposes exactly into news with fixed parameters

## An open question

- Can a nowcasting model beat a simple AR for Romanian GDP growth in a statistically significant way, and which releases carry the information?
  - ▶ formal:  $H_0$ : equal MSE of the DFM and AR nowcasts at the end of month 3 (Diebold–Mariano, [Chapter 1](#)), pre-registered sample, models and lags
  - ▶ falsified by a rejection on quarters the model never saw, with real-time vintages of GDP
- Why it matters: fiscal and monetary decisions in 2025–2026 were taken with GDP known only weeks after each quarter
  - ▶ literature to start from: [Giannone, Reichlin and Small \(2008\)](#), [Bańbura and Modugno \(2014\)](#), [Bańbura, Giannone, Modugno and Reichlin \(2013\)](#), [Giannone, Lenza and Primiceri \(2021\)](#), [Schorfheide and Song \(2015\)](#)

## The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - ▶ **literature:** List peer-reviewed papers that nowcast GDP of Central and Eastern European countries with factor models or MIDAS; give DOIs. Then check every DOI on Crossref
  - ▶ **hypothesis:** Which Romanian monthly releases should carry news about GDP, and in which month of the quarter?
  - ▶ **code and replication:** ask for a Kalman filter with missing data, then reproduce a known number first (the news sum of this lecture)
  - ▶ **robustness and critique:** Act as a hostile referee of a nowcasting paper: list the ways the pseudo-real-time design could flatter the model.
- Report: what was asked, what was kept, what was rejected (AI\_USE.md, AI\_ERRORS.md)

## What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The information set at each date contains only data published by then; current-vintage data overstate accuracy
- ▣ Hyperparameters, factors and lags are chosen inside each estimation window, never on the evaluation sample
- ▣ The evaluation period and the treatment of 2020 are fixed before the results; all variants are reported
- ▣ A claim of “better nowcasts” is backed by a test, not by a lower RMSE alone

## Mini-case: how robust is one ranking?



- RMSE of the DFM nowcast relative to the AR at the end of month 3, 2013–2026, across factors, sample start, release lags and the treatment of 2020
- 16 variants range from 0.71 to 1.03; the DFM wins in 14 of them: an AI summary that reports the best variant as “the” result is wrong

## Project idea

- ▣ **A real-time nowcasting model for Romanian GDP:** replicate first, then extend
  - ▶ replicate: the news decomposition of this lecture ( $-0.51$  pp between August and September 2026) and the BGR table for 1971–2003
  - ▶ extend: real release dates (INS calendar), GDP vintages saved from now on, blocks of factors for hard and soft data, a mixed-frequency BVAR, density nowcasts
  - ▶ pre-register: target, quarters, models, information sets, the treatment of 2020 and the Diebold–Mariano test
- ▣ Deliverables follow the course rules: repository, report, AI\_USE.md, AI\_ERRORS.md, oral defence

## Key takeaways

- ▣ Many series and few observations: shrink (priors) or compress (factors)
- ▣ The Minnesota prior in conjugate form gives closed-form posteriors, exact draws and a marginal likelihood that chooses the shrinkage
- ▣ Bigger systems need tighter priors; then they forecast better and give more plausible responses
- ▣ Principal components estimate the factor space; Bai–Ng chooses its dimension; FAVAR and DFM put it to work
- ▣ Nowcasting is Kalman filtering with a ragged edge; every revision is a sum of news

## Self-assessment

### Questions

- ▣ What happens to the Minnesota posterior mean when  $\lambda \rightarrow 0$ ?
- ▣ Why does the natural conjugate prior force  $\vartheta = 1$ ?
- ▣ Which prior allows cointegration: sum-of-coefficients or initial observation?
- ▣ Why can PCA identify only the factor space?
- ▣ Why is a large but expected release not news?

### Next:

- ▣ State space models and Bayesian filtering
- ▣ The Kalman filter and smoother in full; EM, particle filters, stochastic volatility

## Appendix: the natural conjugate posterior

- ▣ Likelihood:  $p(Y | B, \Sigma) \propto |\Sigma|^{-T/2} \exp\{-\frac{1}{2}\text{tr}[\Sigma^{-1}(Y - XB)'(Y - XB)]\}$
- ▣ Prior:  $|\Sigma|^{-k/2} \exp\{-\frac{1}{2}\text{tr}[\Sigma^{-1}(B - B_0)'\Omega_0^{-1}(B - B_0)]\} \times |\Sigma|^{-(d+n+1)/2} \exp\{-\frac{1}{2}\text{tr}(\Sigma^{-1}\Psi)\}$
- ▣ Complete the square in  $B$ :  

$$(Y - XB)'(Y - XB) + (B - B_0)'\Omega_0^{-1}(B - B_0) = (B - \bar{B})'\bar{\Omega}^{-1}(B - \bar{B}) + \hat{U}'\hat{U} + (\bar{B} - B_0)'\Omega_0^{-1}(\bar{B} - B_0)$$
- ▣ Hence  $B | \Sigma, Y \sim MN(\bar{B}, \bar{\Omega}, \Sigma)$  and  $\Sigma | Y \sim IW(\bar{\Psi}, T + d)$ : the same families as the prior

## Appendix: the marginal likelihood in closed form

- Integrating  $B$  and  $\Sigma$  out (GLP, appendix):  $p(Y) = \pi^{-nT/2} \frac{\Gamma_n(\frac{T+d}{2})}{\Gamma_n(\frac{d}{2})} |\Omega_0|^{-\frac{n}{2}} |X'X + \Omega_0^{-1}|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} |\bar{\Psi}|^{-\frac{T+d}{2}}$
- $|\Omega_0| |X'X + \Omega_0^{-1}| = |I + \Omega_0 X'X|$ : the complexity penalty, growing with  $\lambda$
- $|\bar{\Psi}|$ : the fit term, falling with  $\lambda$ ; the optimum balances the two
- With dummies:  $\ln p(Y | \gamma) = \ln p(Y, Y_d | \gamma) - \ln p(Y_d | \gamma)$ , each by the formula above

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